

Brutal Practice Paper B45 - Respiration, Reproduction in Plants, Simple Equations & Data Handling

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A person breathes 15 times per minute at rest and 45 times per minute while running. In one hour made of 20 minutes running and 40 minutes resting, how many breaths are taken in all? [\[Respiration in Organisms\]](#)
 - (A) 1200 breaths
 - (B) 1050 breaths
 - (C) 1500 breaths
 - (D) 1800 breaths
 - (E) 600 breaths
 - (F) 2700 breaths
2. Yeast reproduces asexually by budding, with one cell splitting into two each cycle. Starting from a single yeast cell, how many cells are there after 5 buddings? [\[Reproduction in Plants\]](#)
 - (A) 25 cells
 - (B) 20 cells
 - (C) 10 cells
 - (D) 16 cells
 - (E) 64 cells
 - (F) 32 cells
 - (G) 5 cells
3. I multiply a number by 4 and subtract 7. The result is the same as 3 times the number plus 1. What is the number? [\[Simple Equations\]](#)
 - (A) 1
 - (B) 2
 - (C) 4
 - (D) 8
 - (E) 9
 - (F) 7
 - (G) -8
 - (H) 6
4. Five breathing-rate readings have a mean of 16 per minute. A sixth reading of 22 per minute is added. What is the new mean of all six readings? [\[Data Handling\]](#)
 - (A) 16
 - (B) 16.5
 - (C) 19
 - (D) 15
 - (E) 22
 - (F) 17

5. Aerobic respiration of one glucose releases about 38 units of energy, while anaerobic respiration of one glucose releases only 2 units. During a sprint a muscle uses 4 glucose aerobically and 6 glucose anaerobically. What total energy is released? [\[Respiration in Organisms\]](#)

- (A) 200 units
- (B) 176 units
- (C) 80 units
- (D) 156 units
- (E) 164 units
- (F) 152 units
- (G) 40 units

6. A potato is a stem tuber that sprouts new plants from its 'eyes' (buds). If one potato has 6 eyes and each grows into a plant, how many new plants come from planting 40 potatoes? [\[Reproduction in Plants\]](#)

- (A) 240 plants
- (B) 246 plants
- (C) 120 plants
- (D) 40 plants
- (E) 200 plants
- (F) 6 plants

7. A plant has a certain number of leaves. After it doubles its leaves and then loses 5, it has 31 leaves. How many leaves did it start with? [\[Simple Equations\]](#)

- (A) 23
- (B) 36
- (C) 26
- (D) 16
- (E) 18
- (F) 9

8. Find the median of this data: 7, 3, 9, 4, 8, 6, 5. [\[Data Handling\]](#)

- (A) 6
- (B) 7
- (C) 8
- (D) 9
- (E) 3
- (F) 4.5
- (G) 5

9. Each breath moves about 500 mL of air in and out, and a resting person breathes 18 times each minute. What volume of air, in litres, is breathed in 10 minutes? [\[Respiration in Organisms\]](#)

- (A) 9 litres
- (B) 900 litres
- (C) 45 litres
- (D) 9000 litres
- (E) 90 litres
- (F) 180 litres
- (G) 50 litres

10. One anther of a flower releases about 2000 pollen grains. A flower has 5 stamens, each with one anther. How many pollen grains can the flower release in total? [\[Reproduction in Plants\]](#)

- (A) 10000 grains
- (B) 2000 grains
- (C) 12000 grains
- (D) 1000 grains
- (E) 5 grains
- (F) 400 grains
- (G) 2005 grains

11. The sum of three consecutive even numbers is 84. What is the smallest of the three numbers? [\[Simple Equations\]](#)

- (A) 84
- (B) 14
- (C) 26
- (D) 30
- (E) 24
- (F) 28

12. Nine test marks are: 5, 7, 7, 8, 5, 7, 9, 5, 7. What is the mode? [\[Data Handling\]](#)

- (A) 9
- (B) 5
- (C) 3
- (D) 8
- (E) 7
- (F) 4
- (G) 6

13. Yeast in bread dough respires without air (anaerobically), releasing carbon dioxide at 30 mL per minute. To rise fully the dough needs 1800 mL of carbon dioxide. How long must it be left to rise? [\[Respiration in Organisms\]](#)

- (A) 600 minutes
- (B) 90 minutes
- (C) 30 minutes
- (D) 6 minutes
- (E) 54 minutes
- (F) 50 minutes
- (G) 60 minutes

14. After pollination, each fertilized ovule becomes one seed. A pea pod's ovary holds 8 ovules, but only 6 of them are fertilized. How many seeds will the pod contain? [\[Reproduction in Plants\]](#)

- (A) 6 seeds
- (B) 8 seeds
- (C) 7 seeds
- (D) 14 seeds
- (E) 48 seeds
- (F) 2 seeds
- (G) 1 seed

- 15.** Right now a man is 4 times as old as his young son; 5 years from today he will be only 3 times as old. What is the son's present age? [\[Simple Equations\]](#)
- (A) 5 years
 - (B) 10 years
 - (C) 8 years
 - (D) 40 years
 - (E) 20 years
 - (F) 15 years
- 16.** Five recorded temperatures are 12, 18, 9, 21 and 15 degrees. What is the range? [\[Data Handling\]](#)
- (A) 21 degrees
 - (B) 12 degrees
 - (C) 15 degrees
 - (D) 30 degrees
 - (E) 18 degrees
 - (F) 13 degrees
 - (G) 9 degrees
- 17.** Inhaled air carries 21 mL of oxygen in every 100 mL, but exhaled air carries only 16 mL in every 100 mL. From 800 mL of air breathed, how much oxygen does the body absorb? [\[Respiration in Organisms\]](#)
- (A) 21 mL
 - (B) 128 mL
 - (C) 40 mL
 - (D) 168 mL
 - (E) 296 mL
 - (F) 5 mL
- 18.** A dandelion seed head holds 200 light, parachute-like seeds. A breeze carries away 65 percent of them to new ground. How many seeds are carried away? [\[Reproduction in Plants\]](#)
- (A) 13 seeds
 - (B) 65 seeds
 - (C) 200 seeds
 - (D) 130 seeds
 - (E) 70 seeds
 - (F) 135 seeds
 - (G) 30 seeds
- 19.** A person takes b breaths per minute at rest and double that while exercising. In a 5-minute spell of 3 minutes rest and 2 minutes exercise, they take 105 breaths. Find b . [\[Simple Equations\]](#)
- (A) 105 breaths per minute
 - (B) 10 breaths per minute
 - (C) 15 breaths per minute
 - (D) 30 breaths per minute
 - (E) 21 breaths per minute
 - (F) 20 breaths per minute
 - (G) 7 breaths per minute

20. The mean of six numbers is 15. Five of them are 12, 18, 10, 20 and 14. What is the sixth number? [\[Data](#)

[Handling\]](#)

- (A) 18
- (B) 14
- (C) 17
- (D) 90
- (E) 16
- (F) 74

21. Insects breathe through tiny pores called spiracles. One insect has 10 pairs of spiracles. A jar holds 8 such insects. How many spiracles are there in all? [\[Respiration in Organisms\]](#)

- (A) 800 spiracles
- (B) 16 spiracles
- (C) 160 spiracles
- (D) 18 spiracles
- (E) 80 spiracles
- (F) 100 spiracles
- (G) 20 spiracles

22. The female part (pistil) of a flower has 3 parts: stigma, style and ovary. Each stamen has 2 parts: anther and filament. A flower has 1 pistil and 6 stamens. How many of these named parts does it have in all? [\[Reproduction in Plants\]](#)

- (A) 12 parts
- (B) 18 parts
- (C) 8 parts
- (D) 9 parts
- (E) 15 parts
- (F) 6 parts
- (G) 21 parts

23. The longer side of a rectangle measures 3 cm more than double its shorter side, and the rectangle's perimeter is 48 cm. How long is the shorter side? [\[Simple Equations\]](#)

- (A) 7 cm
- (B) 8 cm
- (C) 14 cm
- (D) 21 cm
- (E) 6 cm
- (F) 10 cm

24. A bag holds 5 red, 3 blue and 4 green marbles. One marble is drawn at random. What is the probability it is green? [\[Data Handling\]](#)

- (A) $\frac{2}{3}$
- (B) $\frac{4}{8}$
- (C) $\frac{1}{4}$
- (D) $\frac{1}{12}$
- (E) $\frac{1}{6}$
- (F) $\frac{1}{3}$
- (G) $\frac{3}{4}$

25. A plant respire all day and night, releasing 4 units of carbon dioxide every hour. During the 14 daylight hours it also takes in 10 units of carbon dioxide every hour for photosynthesis. Over a full 24-hour day, what is the net change in carbon dioxide? [\[Respiration in Organisms\]](#)

- (A) net releases 56 units
- (B) net absorbs 140 units
- (C) net releases 44 units
- (D) net absorbs 4 units
- (E) net absorbs 96 units
- (F) net absorbs 44 units
- (G) net releases 96 units

26. A fungus spore case releases 5000 spores, but only 2 percent of them land on a suitable damp spot and grow. How many spores grow into new fungi? [\[Reproduction in Plants\]](#)

- (A) 100 spores
- (B) 5000 spores
- (C) 50 spores
- (D) 1000 spores
- (E) 200 spores
- (F) 2500 spores

27. Five identical pens and one Rs 20 notebook together cost Rs 95. What is the price of one pen? [\[Simple Equations\]](#)

- (A) Rs 16
- (B) Rs 23
- (C) Rs 75
- (D) Rs 14
- (E) Rs 19
- (F) Rs 20
- (G) Rs 15

28. A bar graph shows seeds that germinated each day: Monday 20, Tuesday 35, Wednesday 15, Thursday 30. What is the mean number germinating per day? [\[Data Handling\]](#)

- (A) 25 seeds
- (B) 22.5 seeds
- (C) 100 seeds
- (D) 30 seeds
- (E) 50 seeds
- (F) 35 seeds
- (G) 20 seeds

29. When you breathe in, the dome-shaped diaphragm flattens once. At a steady 16 breaths per minute, how many times does the diaphragm flatten in one full hour? [\[Respiration in Organisms\]](#)

- (A) 32 times
- (B) 1920 times
- (C) 96 times
- (D) 960 times
- (E) 16 times
- (F) 480 times
- (G) 9600 times

30. In a garden of 20 flowers, 35 percent are cross-pollinated by visiting bees and the rest are self-pollinated. How many flowers are self-pollinated? [\[Reproduction in Plants\]](#)

- (A) 14 flowers
- (B) 20 flowers
- (C) 7 flowers
- (D) 6 flowers
- (E) 13 flowers
- (F) 35 flowers

31. Half of a number added to one-third of the same number gives 25. What is the number? [\[Simple Equations\]](#)

- (A) 6
- (B) 15
- (C) 50
- (D) 60
- (E) 36
- (F) 25
- (G) 30

32. Class A has 20 students with a mean score of 60, and Class B has 30 students with a mean score of 50. What is the mean score of all 50 students together? [\[Data Handling\]](#)

- (A) 52
- (B) 50
- (C) 55
- (D) 54
- (E) 110
- (F) 56

33. An earthworm breathes through its moist skin. Its body holds 30 mL of water and it loses 2 mL of water every hour in dry air. After how many hours has it lost one-third of its water? [\[Respiration in Organisms\]](#)

- (A) 20 hours
- (B) 15 hours
- (C) 3 hours
- (D) 45 hours
- (E) 7 hours
- (F) 5 hours
- (G) 10 hours

34. One pollen grain can fertilize one ovule, but only 1 in every 5 pollen grains succeeds in reaching an ovule. To fertilize all 50 ovules in an ovary, how many pollen grains are needed? [\[Reproduction in Plants\]](#)

- (A) 100 grains
- (B) 500 grains
- (C) 55 grains
- (D) 10 grains
- (E) 45 grains
- (F) 250 grains

35. Solve for x : $3(x - 4) = x + 2$. [\[Simple Equations\]](#)

- (A) 5
- (B) 14
- (C) 6
- (D) 3.5
- (E) -7
- (F) 7
- (G) 10

36. Find the median of: 10, 14, 8, 12, 20, 16. [\[Data Handling\]](#)

- (A) 16
- (B) 12
- (C) 14
- (D) 12.5
- (E) 13.5
- (F) 13
- (G) 11

37. When muscles run short of oxygen they respire anaerobically, building up lactic acid that causes cramp. If lactic acid builds at 3 units per minute and cramp starts at 24 units, after how many minutes of hard effort does cramp begin? [\[Respiration in Organisms\]](#)

- (A) 27 minutes
- (B) 6 minutes
- (C) 21 minutes
- (D) 72 minutes
- (E) 4 minutes
- (F) 8 minutes

38. In Bryophyllum, tiny new plants grow from buds in the notches along the leaf edge. If one leaf has 12 such notches and 3 leaves are placed on damp soil, how many new plantlets can grow? [\[Reproduction in Plants\]](#)

- (A) 24 plantlets
- (B) 4 plantlets
- (C) 36 plantlets
- (D) 9 plantlets
- (E) 48 plantlets
- (F) 12 plantlets
- (G) 15 plantlets

39. Two identical seed pods together hold 6 more seeds than the number in one pod plus 14. Written as an equation, $2s + 6 = s + 14$. How many seeds are in one pod? [\[Simple Equations\]](#)

- (A) 10 seeds
- (B) 4 seeds
- (C) 6 seeds
- (D) 20 seeds
- (E) 14 seeds
- (F) 8 seeds

40. A fair die is rolled once. What is the probability of NOT getting a 4? [\[Data Handling\]](#)

- (A) $\frac{2}{3}$
- (B) $\frac{1}{6}$
- (C) $\frac{5}{6}$
- (D) $\frac{1}{3}$
- (E) $\frac{4}{6}$
- (F) $\frac{1}{2}$
- (G) 1

41. Fully respiring 1 gram of glucose releases about 16 units of energy. How many grams of glucose must be respired to release 240 units of energy? [\[Respiration in Organisms\]](#)

- (A) 256 grams
- (B) 15 grams
- (C) 224 grams
- (D) 240 grams
- (E) 3840 grams
- (F) 30 grams
- (G) 16 grams

42. A mango tree produces 5000 flowers in a season, but only 4 percent of them are fertilized and set fruit. How many mangoes does the tree bear? [\[Reproduction in Plants\]](#)

- (A) 20 fruits
- (B) 200 fruits
- (C) 1250 fruits
- (D) 5000 fruits
- (E) 500 fruits
- (F) 100 fruits

43. The mean of five readings 10, 14, 9, 15 and x is 12. Find x. [\[Simple Equations\]](#)

- (A) 11
- (B) 15
- (C) 48
- (D) 60
- (E) 13
- (F) 12
- (G) 10

44. The mean of 10 numbers is 50. When one number is removed, the mean of the remaining 9 numbers is 52. What was the number that was removed? [\[Data Handling\]](#)

- (A) 42
- (B) 2
- (C) 52
- (D) 18
- (E) 50
- (F) 48
- (G) 32

45. Fish breathe by passing water over their gills, but water holds little oxygen: 1 litre of water has only 8 mL of dissolved oxygen. A fish needs 40 mL of oxygen each hour. How many litres of water must pass over its gills each hour? [\[Respiration in Organisms\]](#)

- (A) 32 litres
- (B) 10 litres
- (C) 8 litres
- (D) 5 litres
- (E) 40 litres
- (F) 320 litres
- (G) 48 litres

46. A seed packet holds 80 seeds and claims an 85 percent germination rate. How many of the seeds are expected to sprout? [\[Reproduction in Plants\]](#)

- (A) 76 seeds
- (B) 68 seeds
- (C) 8 seeds
- (D) 60 seeds
- (E) 12 seeds
- (F) 85 seeds
- (G) 80 seeds

47. In a triangle two angles are equal, and the third is 30 degrees more than each of them. How big is each of the two equal angles? (Angles of a triangle add to 180 degrees.) [\[Simple Equations\]](#)

- (A) 40 degrees
- (B) 45 degrees
- (C) 30 degrees
- (D) 80 degrees
- (E) 60 degrees
- (F) 50 degrees
- (G) 75 degrees

48. A frequency table shows: value 2 occurs 3 times, value 4 occurs 5 times, value 6 occurs 2 times. What is the mean of all the data? [\[Data Handling\]](#)

- (A) 4.5
- (B) 38
- (C) 3
- (D) 4
- (E) 3.5
- (F) 12.67
- (G) 3.8

49. A new-born baby breathes about 40 times per minute, while an adult breathes about 16 times per minute. Over 5 minutes, how many more breaths does the baby take than the adult? [\[Respiration in Organisms\]](#)

- (A) 24 breaths
- (B) 120 breaths
- (C) 100 breaths
- (D) 280 breaths
- (E) 56 breaths
- (F) 200 breaths
- (G) 80 breaths

50. A bee carrying pollen visits 15 flowers on each trip and makes 8 trips in a morning. How many flower visits, helping cross-pollination, does it make in all? [\[Reproduction in Plants\]](#)

- (A) 8 flowers
- (B) 100 flowers
- (C) 240 flowers
- (D) 60 flowers
- (E) 23 flowers
- (F) 15 flowers
- (G) 120 flowers

51. Five times a number, decreased by 9, equals three times the number increased by 7. Find the number.

[\[Simple Equations\]](#)

- (A) 16
- (B) 1
- (C) 5
- (D) 4
- (E) -8
- (F) 8
- (G) 2

52. In a pictograph, one symbol stands for 8 plants. A row shows 5 and a half symbols. How many plants does the row represent? [\[Data Handling\]](#)

- (A) 88 plants
- (B) 36 plants
- (C) 44 plants
- (D) 48 plants
- (E) 40 plants
- (F) 13.5 plants
- (G) 5.5 plants

53. Exhaled air contains about 4 percent carbon dioxide, while inhaled air contains only about 0.04 percent. How many times more carbon dioxide is in the breath we breathe out? [\[Respiration in Organisms\]](#)

- (A) 96 times
- (B) 100 times
- (C) 1000 times
- (D) 25 times
- (E) 4 times
- (F) 40 times
- (G) 10 times

- 54.** A gardener plants 25 rose cuttings to grow new plants vegetatively. Only 60 percent of them root and survive. How many new rose plants are obtained? [\[Reproduction in Plants\]](#)
- (A) 12 cuttings
 - (B) 25 cuttings
 - (C) 20 cuttings
 - (D) 60 cuttings
 - (E) 10 cuttings
 - (F) 15 cuttings
 - (G) 40 cuttings
- 55.** A box holds Rs 2 coins and Rs 5 coins, 30 coins in all, worth Rs 99 together. How many Rs 5 coins are there? [\[Simple Equations\]](#)
- (A) 17 coins
 - (B) 21 coins
 - (C) 13 coins
 - (D) 30 coins
 - (E) 15 coins
 - (F) 9 coins
 - (G) 11 coins
- 56.** From a box of tickets numbered 1 to 20, one ticket is picked at random. Find the probability that its number is a multiple of 5. [\[Data Handling\]](#)
- (A) $\frac{2}{5}$
 - (B) $\frac{4}{5}$
 - (C) $\frac{1}{10}$
 - (D) $\frac{1}{20}$
 - (E) $\frac{1}{5}$
 - (F) $\frac{3}{20}$
- 57.** In an insect's breathing tubes, each trachea splits into 2 smaller tubes at every branch point. Starting from 1 main trachea, how many tubes are there after 4 successive branchings? [\[Respiration in Organisms\]](#)
- (A) 4 tubes
 - (B) 24 tubes
 - (C) 8 tubes
 - (D) 16 tubes
 - (E) 64 tubes
 - (F) 12 tubes
 - (G) 32 tubes
- 58.** A pollen grain is too small to see. Under a microscope that magnifies 10 to the power 2 times, a grain appears 5 mm wide. What is its real width? [\[Reproduction in Plants\]](#)
- (A) 0.5 mm
 - (B) 50 mm
 - (C) 0.05 mm
 - (D) 5 mm
 - (E) 0.005 mm
 - (F) 500 mm

- 59.** A car travels x km in the first hour, twice as far in the second hour, and 10 km less than the first hour in the third hour. The three hours cover 90 km. How far is the first hour? [\[Simple Equations\]](#)
- (A) 20 km
 - (B) 35 km
 - (C) 45 km
 - (D) 22.5 km
 - (E) 15 km
 - (F) 30 km
 - (G) 25 km
- 60.** Five temperature readings are -3, 5, -8, 2 and 7 degrees. What is the range? [\[Data Handling\]](#)
- (A) 7 degrees
 - (B) 15 degrees
 - (C) 10 degrees
 - (D) 11 degrees
 - (E) 2 degrees
 - (F) 8 degrees
 - (G) 12 degrees
- 61.** After a sprint a runner pays back an 'oxygen debt' by breathing fast: 35 breaths per minute for 6 minutes, instead of the normal 15 per minute. How many extra breaths above normal are taken during this recovery? [\[Respiration in Organisms\]](#)
- (A) 60 breaths
 - (B) 200 breaths
 - (C) 210 breaths
 - (D) 35 breaths
 - (E) 120 breaths
 - (F) 105 breaths
- 62.** A balsam pod bursts open to fling its seeds away. Each pod holds 9 seeds, and one plant has 30 ripe pods. How many seeds can the plant scatter by this bursting? [\[Reproduction in Plants\]](#)
- (A) 90 seeds
 - (B) 270 seeds
 - (C) 9 seeds
 - (D) 300 seeds
 - (E) 30 seeds
 - (F) 39 seeds
- 63.** After growing 20 percent taller, a plant is 96 cm high. How tall was it before the growth? [\[Simple Equations\]](#)
- (A) 80 cm
 - (B) 120 cm
 - (C) 100 cm
 - (D) 116 cm
 - (E) 19.2 cm
 - (F) 76.8 cm

64. What is the mean of the first five multiples of 4? [\[Data Handling\]](#)

- (A) 60
- (B) 12
- (C) 15
- (D) 8
- (E) 10
- (F) 20
- (G) 4

65. A sealed glass jar in sunlight holds a plant that releases 18 mL of oxygen per hour by net photosynthesis, and a small animal that uses 6 mL of oxygen per hour to respire. By how much does the oxygen in the jar change each hour? [\[Respiration in Organisms\]](#)

- (A) increases by 12 mL
- (B) increases by 24 mL
- (C) decreases by 12 mL
- (D) increases by 9 mL
- (E) increases by 3 mL
- (F) increases by 6 mL
- (G) increases by 18 mL

66. Coconuts float on water and drift to new shores to grow. Of 250 coconuts that drop near a beach, 12 percent float away and take root elsewhere. How many start new palms on other shores? [\[Reproduction in Plants\]](#)

- (A) 25 coconuts
- (B) 220 coconuts
- (C) 36 coconuts
- (D) 3 coconuts
- (E) 30 coconuts
- (F) 60 coconuts
- (G) 12 coconuts

67. Solve for x: $7x - 5 = 4x + 16$. [\[Simple Equations\]](#)

- (A) 5
- (B) 6
- (C) 3
- (D) 7
- (E) 21
- (F) 9
- (G) 11

68. Find the median of: 5, 5, 5, 9, 12, 12, 18. [\[Data Handling\]](#)

- (A) 6
- (B) 9.4
- (C) 8
- (D) 12
- (E) 5
- (F) 66
- (G) 9

69. A diver's air tank holds 2000 litres of air, and the diver breathes 25 litres of air each minute. For how many minutes will the tank last? [\[Respiration in Organisms\]](#)

- (A) 800 minutes
- (B) 40 minutes
- (C) 75 minutes
- (D) 80 minutes
- (E) 50000 minutes
- (F) 100 minutes
- (G) 8 minutes

70. Insect-pollinated flowers are bright and scented, while wind-pollinated ones are dull. In a field of 40 flowers, $\frac{3}{8}$ are bright and scented (insect-pollinated) and the rest are wind-pollinated. How many are wind-pollinated? [\[Reproduction in Plants\]](#)

- (A) 3 flowers
- (B) 8 flowers
- (C) 10 flowers
- (D) 5 flowers
- (E) 24 flowers
- (F) 15 flowers
- (G) 25 flowers

71. A teacher gives each student 4 worksheets and keeps 7 spare. She uses 95 worksheets in all. How many students are there? [\[Simple Equations\]](#)

- (A) 23 students
- (B) 21 students
- (C) 88 students
- (D) 25 students
- (E) 22 students
- (F) 24 students
- (G) 20 students

72. A student scored a mean of 80 over 4 tests. What must the score be in a 5th test to raise the mean to 82 over all 5 tests? [\[Data Handling\]](#)

- (A) 100 marks
- (B) 84 marks
- (C) 80 marks
- (D) 88 marks
- (E) 90 marks
- (F) 82 marks

73. In respiration, every carbon atom in glucose leaves as one molecule of carbon dioxide. Glucose has 6 carbon atoms. How many carbon dioxide molecules are produced from 7 glucose molecules? [\[Respiration in Organisms\]](#)

- (A) 6 molecules
- (B) 7 molecules
- (C) 36 molecules
- (D) 12 molecules
- (E) 49 molecules
- (F) 13 molecules
- (G) 42 molecules

74. A bean plant flowers 90 days after a seed is sown, then takes another 40 days to ripen its own seeds. Counting from sowing, after how many days are the new seeds ready? [\[Reproduction in Plants\]](#)

- (A) 120 days
- (B) 3600 days
- (C) 90 days
- (D) 50 days
- (E) 230 days
- (F) 130 days

75. A tree bears a certain number of fruits this year. Next year it bears 12 fewer than three times as many, which is 60 fruits. How many fruits did it bear this year? [\[Simple Equations\]](#)

- (A) 12 fruits
- (B) 48 fruits
- (C) 36 fruits
- (D) 72 fruits
- (E) 16 fruits
- (F) 20 fruits
- (G) 24 fruits

76. A spinner has 8 equal sectors: 3 red, 2 blue, 2 yellow and 1 green. What is the probability of landing on red or green? [\[Data Handling\]](#)

- (A) $\frac{7}{8}$
- (B) $\frac{3}{4}$
- (C) $\frac{1}{2}$
- (D) $\frac{5}{8}$
- (E) $\frac{3}{8}$
- (F) $\frac{1}{4}$

77. A frog on land gets its oxygen partly through its lungs and partly through its skin: lungs supply 75 percent and skin supplies 25 percent. If the frog needs 80 mL of oxygen per hour, how much comes through the skin? [\[Respiration in Organisms\]](#)

- (A) 80 mL
- (B) 25 mL
- (C) 60 mL
- (D) 75 mL
- (E) 20 mL
- (F) 40 mL

- 78.** An apple holds 5 seeds. Suppose all 5 grow into trees, and each tree later bears 30 apples. How many apples does this next generation produce in total? [\[Reproduction in Plants\]](#)
- (A) 6 apples
 - (B) 30 apples
 - (C) 35 apples
 - (D) 150 apples
 - (E) 75 apples
 - (F) 5 apples
- 79.** An isosceles triangle has a perimeter of 40 cm and a base of 12 cm. How long is each of the two equal sides? [\[Simple Equations\]](#)
- (A) 26 cm
 - (B) 14 cm
 - (C) 20 cm
 - (D) 28 cm
 - (E) 16 cm
 - (F) 13 cm
 - (G) 12 cm
- 80.** A bar graph of favourite fruits among 50 students shows: apple 15, banana 20, mango 10, others 5. What percentage chose banana? [\[Data Handling\]](#)
- (A) 10%
 - (B) 40%
 - (C) 45%
 - (D) 20%
 - (E) 50%
 - (F) 30%
 - (G) 35%
- 81.** A leaf carries about 250 stomata in each square millimetre, the pores used for gas exchange. A small leaf measures 8 square millimetres. About how many stomata does it have? [\[Respiration in Organisms\]](#)
- (A) 200 stomata
 - (B) 258 stomata
 - (C) 1000 stomata
 - (D) 2000 stomata
 - (E) 500 stomata
 - (F) 250 stomata
 - (G) 20000 stomata
- 82.** A lily flower has 6 stamens, and a buttercup has 10. A bouquet holds 4 lilies and 4 buttercups. How many anthers (one per stamen) are in the bouquet altogether? [\[Reproduction in Plants\]](#)
- (A) 56 anthers
 - (B) 64 anthers
 - (C) 32 anthers
 - (D) 24 anthers
 - (E) 16 anthers
 - (F) 80 anthers
 - (G) 40 anthers

83. The sum of a whole number and the next number after it is 47. What is the smaller number? [\[Simple Equations\]](#)

- (A) 22
- (B) 11.5
- (C) 47
- (D) 24
- (E) 23
- (F) 46
- (G) 25

84. Six plants have a mean height of 24 cm. A seventh plant, 38 cm tall, is added. What is the new mean height? [\[Data Handling\]](#)

- (A) 25 cm
- (B) 26 cm
- (C) 38 cm
- (D) 30 cm
- (E) 24 cm
- (F) 31 cm

85. In anaerobic respiration, yeast turns glucose into alcohol and carbon dioxide. If each kilogram of glucose yields 0.5 kg of alcohol, how much alcohol forms when 50 kg of glucose ferments? [\[Respiration in Organisms\]](#)

- (A) 2.5 kg
- (B) 5 kg
- (C) 50 kg
- (D) 12.5 kg
- (E) 25 kg
- (F) 75 kg

86. A strawberry plant spreads by runners, each making identical daughter plants. If every plant makes 4 new plants per year, and you start with 1 plant, how many plants are there after 2 years (including the original)? [\[Reproduction in Plants\]](#)

- (A) 21 plants
- (B) 25 plants
- (C) 9 plants
- (D) 16 plants
- (E) 24 plants
- (F) 20 plants
- (G) 8 plants

87. A girl spends half of her money, then Rs 30 more, and is left with Rs 20. How much money did she start with? [\[Simple Equations\]](#)

- (A) Rs 20
- (B) Rs 110
- (C) Rs 100
- (D) Rs 50
- (E) Rs 80
- (F) Rs 40
- (G) Rs 60

88. Shoe sizes sold in a day were: 6, 7, 7, 8, 7, 9, 6, 7. Which size should the shop stock the most of (the mode)? [\[Data Handling\]](#)

- (A) size 8
- (B) size 7.25
- (C) size 6.5
- (D) size 9
- (E) size 6
- (F) size 7
- (G) size 5

89. Aerobic respiration of glucose releases 38 units of energy, while anaerobic respiration of the same glucose releases only 2 units. How many times more energy does aerobic respiration give? [\[Respiration in Organisms\]](#)

- (A) 18 times
- (B) 40 times
- (C) 19 times
- (D) 36 times
- (E) 2 times
- (F) 20 times
- (G) 76 times

90. A sunflower head carries 1000 seeds that together weigh 60 grams. What is the average mass of one seed, in milligrams? (1 gram = 1000 milligrams.) [\[Reproduction in Plants\]](#)

- (A) 600 mg
- (B) 0.06 mg
- (C) 6 mg
- (D) 6000 mg
- (E) 60 mg
- (F) 0.6 mg
- (G) 16 mg

91. A number is as much above 20 as it is below 80. What is the number? [\[Simple Equations\]](#)

- (A) 25
- (B) 50
- (C) 40
- (D) 45
- (E) 30
- (F) 60
- (G) 100

92. In a class of 36 students, one-third play cricket and one-quarter play football. How many play neither of these two games? [\[Data Handling\]](#)

- (A) 24 students
- (B) 18 students
- (C) 21 students
- (D) 15 students
- (E) 3 students
- (F) 12 students

93. At rest a person breathes 15 times per minute. After climbing stairs, the breathing rate rises by 60 percent. What is the new breathing rate? [\[Respiration in Organisms\]](#)

- (A) 24 breaths per minute
- (B) 21 breaths per minute
- (C) 75 breaths per minute
- (D) 25 breaths per minute
- (E) 60 breaths per minute
- (F) 9 breaths per minute
- (G) 16 breaths per minute

94. A field is sown with 5000 seeds. 90 percent of them germinate, and of those that germinate, 80 percent survive to become adult plants. How many adult plants are there? [\[Reproduction in Plants\]](#)

- (A) 4000 plants
- (B) 3240 plants
- (C) 3500 plants
- (D) 4500 plants
- (E) 3600 plants
- (F) 900 plants

95. Aerobic respiration releases 6 units more than 16 times the anaerobic energy e , and this total is 38 units. Find the anaerobic energy e . [\[Simple Equations\]](#)

- (A) 3 units
- (B) 2.75 units
- (C) 4 units
- (D) 32 units
- (E) 44 units
- (F) 2 units
- (G) 2.5 units

96. A set of data is 4, 5, 6, 7 and 100. What is the median? [\[Data Handling\]](#)

- (A) 5
- (B) 24.4
- (C) 6
- (D) 7
- (E) 12.2
- (F) 100

97. A person at rest breathes 16 times per minute. How many breaths are taken in one full day? [\[Respiration in Organisms\]](#)

- (A) 384 breaths
- (B) 11520 breaths
- (C) 960 breaths
- (D) 23040 breaths
- (E) 2304 breaths
- (F) 46080 breaths
- (G) 1440 breaths

98. A flower makes 8000 pollen grains and has 40 ovules. On average, how many pollen grains are produced for each ovule? [\[Reproduction in Plants\]](#)

- (A) 100 grains per ovule
- (B) 5 grains per ovule
- (C) 160 grains per ovule
- (D) 320000 grains per ovule
- (E) 40 grains per ovule
- (F) 200 grains per ovule
- (G) 320 grains per ovule

99. A rectangle's length is 5 times its width, and its perimeter is 60 cm. Find the width. [\[Simple Equations\]](#)

- (A) 5 cm
- (B) 15 cm
- (C) 30 cm
- (D) 25 cm
- (E) 10 cm
- (F) 12 cm

100. Rainfall over four weeks was 20 mm, 35 mm, an unknown amount, and 40 mm, totalling 130 mm. How much rain fell in the third week? [\[Data Handling\]](#)

- (A) 35 mm
- (B) 32.5 mm
- (C) 30 mm
- (D) 95 mm
- (E) 25 mm
- (F) 130 mm